EMFT (UE24EE242B)

Electrostatics Unit 1

QB

Electric Field due to continuous distribution of charges:

1. Obtain the expression for Electric field due to a Finite line charge along Z axis at a point p and hence extend the same for an Infinite line charge.

Numerical on Electric Field due to continuous distribution of charges:

1.

The finite sheet $0 \le x \le 1$, $0 \le y \le 1$ on the $z = 0$ plane has a charge density $\rho_S = xy(x^2 + y^2 + 25)^{3/2}$ nC/m². Find

(a) The total charge on the sheet

(b) The electric field at (0, 0, 5)

(c) The force experienced by a -1 mC charge located at (0, 0, 5)

2.

A square plate described by $-2 \le x \le 2$, $-2 \le y \le 2$, $z = 0$ carries a charge $12\,|y|$ mC/m². Find the total charge on the plate and the electric field intensity at (0, 0, 10).

3.

Planes $x = 2$ and $y = -3$, respectively, carry charges 10 nC/m² and 15 nC/m². If the line $x = 0$, $z = 2$ carries charge 10π nC/m, calculate **E** at (1, 1, -1) due to the three charge distributions.

4.

Planes $x = 2$ and $y = -3$, respectively, carry charges 10 nC/m² and 15 nC/m². if the line $x = 0$, $z = 2$ is rotated through 90° about the point (0, 2, 2) so that it becomes $x = 0$, $y = 2$, find **E** at (1, 1, -1).

5.

Determine the total charge

(a) On line $0 < x < 5$ m if $\rho_L = 12x^2$ mC/m

(b) On the cylinder $\rho = 3$, $0 < z < 4$ m if $\rho_S = \rho z^2$ nC/m²

(c) Within the sphere $r = 4$ m if $\rho_v = \dfrac{10}{r \sin\theta}$ C/m³

6.

A circular ring of radius a carries a uniform charge ρ_L C/m and is placed on the xy-plane with axis the same as the z-axis.

(a) Show that

$$\mathbf{E}(0, 0, h) = \frac{\rho_L a h}{2\varepsilon_o[h^2 + a^2]^{3/2}} \mathbf{a}_z$$

(b) What values of h gives the maximum value of $\mathbf{E}$?

(c) If the total charge on the ring is Q, find $\mathbf{E}$ as $a \to 0$.

7.

A circular disk of radius a is uniformly charged with ρ_S C/m^2. If the disk lies on the $z = 0$ plane with its axis along the z-axis,

(a) Show that at point $(0, 0, h)$

$$\mathbf{E} = \frac{\rho_S}{2\varepsilon_o}\left\{1 - \frac{h}{[h^2 + a^2]^{1/2}}\right\}\mathbf{a}_z$$

(b) From this, derive the $\mathbf{E}$ field due to an infinite sheet of charge on the $z = 0$ plane.